

Meridian Sorbent Technologies

Materials Characterisation Laboratory

Thermal screening note: composite sorbent programme

Note ref. MST-TGA-2018-09 | issued 2 October 2018 | supersedes MST-TGA-2018-04

Runs covered

Four thermogravimetric runs were made on the September samples under a single method: a 10 °C/min ramp to 500 °C in a platinum pan under nitrogen, balance purge 10 mL/min and sample purge 90 mL/min. They are serials 828 and 829 on the composite, before and after a phosphate loading cycle, and serials 830 and 831 on the two precursors.

Peak selection

The sorbents in this programme hold sorbed and structurally bound water, and every run in the set opens with a mass-loss step that is that water leaving the sample rather than the material breaking down. That step is complete by 150 °C in all four runs, so a primary degradation peak is read above 150 °C and the opening step is left out of the selection. The peak of record for a run is the largest rate of mass loss reached above that point, quoted with the temperature at which it occurs.

Reading of record

The overlay circulated with this note, DTG_Thermogram.png, is the laboratory's reviewed reading of these four runs and is the figure the programme quotes from. The instrument exports are supplied alongside it so that the underlying traces are available to anyone who needs to work with them. Where an export and the overlay disagree on a peak, the overlay is the reading that stands.

Handling limits

A handling or regeneration ceiling is quoted on the material as recovered, and is the temperature at which the sample has lost five per cent of the mass it still holds at 150 °C.

Use of these runs

The runs are screening data. They characterise thermal behaviour under a nitrogen ramp and say nothing on their own about performance in service, sorption capacity or regeneration yield.